

The economics of bicycles for the mind:

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$$V = \frac{\Delta^2}{4}, \quad H = kT\alpha^2 s, \quad z = \frac{\pi}{s}$$

$$V(\theta) = H + \theta z$$

$$\begin{aligned} \bullet a_2 &= E[\alpha^2] & \bullet a_4 &= E[\alpha^4] \\ \bullet g_1 &= E[\pi] & \bullet g_2 &= E[\pi^2] \\ \bullet s_1 &= E[s] & \bullet s_2 &= E[s^2] \\ \bullet r_1 &= E[1/s] & \bullet r_2 &= E[1/s^2] \end{aligned}$$

donde T, α, s son independientes, solución interior

A.) Esperanza

$$E[V] = E[H] + \theta E[z]$$

usando independencia

$$E[H] = k E[T] E[\alpha^2] E[s] = k g_1 a_2 s_1$$

$$E[z] = E[\pi] E[1/s] = g_1 r_1$$

$$E[V] = k g_1 \cdot a_2 \cdot s_1 + \theta g_1 r_1 \quad \text{Q}$$

B.) Identidad de la variancia

$$V - E[V] = (H - E[H]) + \theta (z - E[z])$$

$$[V - E[V]]^2 = [(H - E[H])]^2 + 2\theta (H - E[H])(z - E[z]) + \theta^2 (z - E[z])^2$$

Tomando esperanzas

$$\text{var}(V) = \text{var}(H) + 2\theta \text{cov}(H, z) + \theta^2 \text{var}(z)$$

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Calculando los coeficientes:

A.) $\text{var}(H)$

$$H^2 = k^2 T^2 \alpha^4 s^2$$

$$E[H^2] = k^2 g_2 a_4 s_2$$

$$\text{var}(H) = k^2 (g_2 a_4 s_2 - g_1^2 a_2^2 s_1^2)$$

B.) $\text{var}(z)$

$$z^2 = \frac{\pi^2}{s^2}, \quad E[z^2] = g_2 r_2$$

$$\text{var}(z) = g_2 r_2 - g_1^2 r_1^2$$

C.) $\text{cov}(H, z)$

$$Hz = (kT\alpha^2 s) \left(\frac{\pi}{s} \right) = kT\alpha^2 \pi$$

$$E[Hz] = k g_2 a_2$$

$$E[H] \cdot E[z] = k a_2 g_1^2 s_1 r_1$$

$$\text{cov}(H, z) = k a_2 (g_2 - g_1^2 s_1 r_1)$$

$\Rightarrow$ donde, debemos sustituir los cuadrados por la identidad de la Hoja 1.

derivadas, FOC y turning point:

$$\frac{\partial \text{var}(V)}{\partial \theta} = 2 \text{cov}(H, z) + 2\theta \text{var}(z)$$

$$\frac{\partial^2 \text{var}(V)}{\partial^2 \theta} = 2 \text{var}(z) \geq 0$$

$$\frac{\partial \text{var}(V)}{\partial \theta} = 2 k a_2 (g_2 - g_1^2 s_1 r_1)$$

$$\text{como } 2 k a_2 > 0, \quad \frac{g_2}{g_1^2} < s_1 r_1 = E[1/s] \dots (30)$$

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$$\begin{aligned} \text{var}(z) &= \frac{g_2}{r_2} - \left(\frac{g_1}{r_1} \right)^2 = \frac{1}{149} - \left(\frac{1}{117} \right)^2 = \frac{1}{149} - \frac{1}{13689} = \frac{94}{13689} \\ E[1/s^2] &= \frac{1}{100} + \frac{1}{149} = \frac{149}{14900} + \frac{100}{14900} = \frac{249}{14900} \\ \text{cov}(H, z) &= \frac{64}{25} \left(1 - \frac{260}{139} \right) = \frac{64}{25} \left(\frac{139 - 260}{139} \right) = \frac{64}{25} \left(\frac{-121}{139} \right) = -\frac{7744}{3475} \\ r_2 &= \frac{1}{16}, \quad a_2 = \frac{1}{16}, \quad k \cdot a_2 = \frac{1}{16} \end{aligned}$$

calculando covarianzas y variancias.

condición (30) se cumple.

entonces: $1 < \frac{12}{17} \cdot \frac{20}{14} = \frac{280}{289}$, de modo que

$$E[s] = \frac{12}{17}, \quad s_2 = \frac{20}{3}, \quad \frac{20}{17} < \frac{20}{3} > \frac{20}{3}$$

$$\Delta = 4, \quad s = \frac{10}{3}, \quad s = 1 \text{ con prob. } 0.5$$

$$\text{sea: } \delta = \frac{1}{2}, \quad \delta_0 = \frac{4}{3}, \quad r = \frac{1}{1}, \quad T = 1, \quad \alpha = \frac{1}{4}$$

Como contraejemplo:

$\therefore (30)$ da $\theta^* > 0$, mas no $\theta^* < 1$.

pendiente evaluada en $\theta^* > 0 \Rightarrow \theta^* < 1 \Rightarrow \text{var}(z) > 0$

$$\text{evaluado en } \theta^*: \frac{\partial \text{var}(V)}{\partial \theta} = 2 \text{var}(z) / (1 - \theta^*)$$

$$\text{entonces: } \frac{\partial \text{var}(V)}{\partial \theta} = 2 \text{var}(z) / (1 - \theta^*)$$

$$\frac{\text{var}(z)}{\text{cov}(H, z)} = \frac{k a_2 (g_2 s_1 r_1 - g_1^2)}{g_2 r_2 - g_1^2 r_1^2}$$

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$$\theta^* = \frac{72/875}{9/146} = \frac{224}{125} = 1.792$$

$\Rightarrow$ la condición (30) se cumple, pero la variancia sigue cayendo en $\theta = 1$.

si partimos de la solución interior

$$c_{\text{int}}^* = \frac{\alpha^2 \Delta^2 s}{4} - \frac{\theta}{s}$$

imponiendo $e \geq 0$:

$$e^* = \max \left\{ 0, \frac{\alpha^2 \Delta^2 s}{4} - \frac{\theta}{s} \right\} \text{ con}$$

$$T = \frac{\alpha^2 \Delta^2 s^2}{4}$$

sustituyendo cada rotación en el objetivo

$$m^*(\theta) = \begin{cases} \frac{\alpha^2 \Delta^2 s}{4} + \frac{\theta}{s}, & \theta \leq T \\ \alpha \Delta \sqrt{\theta}, & \theta \geq T \end{cases}$$

verificamos continuidad en T : ambas ramas valen

$$\frac{\alpha^2 \Delta^2 s}{4}$$

verificamos que las primeras derivadas coincidan

$$\frac{1}{s} = \frac{\alpha \Delta}{2\sqrt{T}}$$

$\Rightarrow$ después de que algunos agentes llegan a la esquina, ya no es $H + \theta z$ para todos: la fórmula cuadrática de la variancia deja de ser global.

$\Rightarrow$ si todos están en la esquina:

$$\text{var}(V(\theta)) = \Delta^2 \theta \text{var}(T\alpha)$$